Answers written in the margins will not be marked.

10. You are given a long conducting wire, a pair of slab-shaped magnets on steel yoke and a light-beam galvanometer for detecting small currents. With the aid of a diagram, describe an experiment to investigate **TWO** factors affecting the e.m.f. induced in a conductor when it moves in a magnetic field. (7 marks)

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(ii) Explain why within the very short duration of lightning an induced current first flows in the coil in a certain direction and then reverses. Your answer should include the directions of the induced current in the coil. (3 marks)

(iii) Among the physical quantities related to lightning, **electric field in the atmosphere**, **lightning current** and **magnetic field due to lightning**, suggest which one can be monitored so as to give fore-warning of lightning. Explain your choice. (2 marks)

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(b) As the first decay in the above chain from U to Th has a half-life much longer than those of subsequent decays, the decay from U-238 to Pb-206 can be simplified to *a single decay* with half-life 4.5×10^9 years:

$$^{238}_{92}\mathrm{U} \xrightarrow[4.5\times 10^9 \text{ years}]{} {}^{206}_{82}\mathrm{Pb}$$

Suppose that a uranium-bearing rock contains only U-238 and no Pb-206 at the time when it was formed long ago by solidification of molten material. In a particular sample of the rock, it is found that the ratio $\dfrac{\text{number of Pb-206 atoms}}{\text{number of U-238 atoms}}$ is $\dfrac{2}{3}$ *at present.*

(i) Estimate the age of the rock. Assume that all Pb-206 atoms come from the decay of U-238 originally present in the sample and ignore the small number of U-238 atoms which have decayed but have not yet become Pb-206. (2 marks)

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(iii) The graph in Figure 9.1 shows how the number of U-238 atoms in the sample varies with time t subsequently while $t = 0$ denotes *the present time*. On Figure 9.1, sketch a graph to show the variation of the number of Pb-206 atoms in the sample with time. (2 marks)

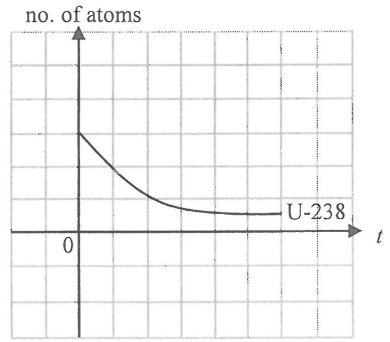

Figure 9.1

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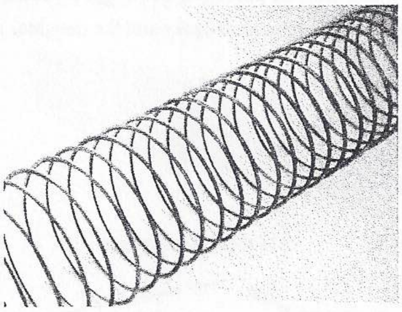
(b) Figure 9.2 shows a metal slinky spring.

Figure 9.2



(i) If a direct current passes through the spring, briefly explain whether the spring will be compressed or stretched due to magnetic force. (2 marks)

(ii) A student suggests that the spring will be compressed and stretched alternately due to magnetic force when an alternating current passes through. Briefly explain why he is wrong. (1 mark)

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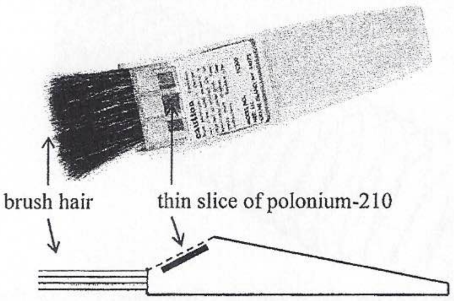
10. Dust may adhere to the surfaces of photos and films due to electrostatic attraction. To remove the dust effectively, a special brush with a thin slice of polonium-210 ($^{210}_{84}\text{Po}$) fixed near the brush hair as shown in Figure 10.1 may be used. Polonium-210 undergoes α decay and the daughter nucleus lead (Pb) is stable.

Figure 10.1



(a) Write a nuclear equation for the decay of polonium-210. (2 marks)

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(b) Briefly explain how the α particles help clean the charged dust. (2 marks)

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(c) Briefly explain why the polonium-210 slice must be fixed near to the brush hair. (1 mark)

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*(d) The manufacturer recommends that the brush should be returned to the factory for replacement of the polonium-210 slice every year. Taking the activity of a newly replaced polonium-210 slice as 1 unit, find its activity after one year (365 days). Given: half-life of polonium-210 is 138 days. (2 marks)

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List of data, formulae and relationships

Data

molar gas constant	$R = 8.31\ \mathrm{J\ mol^{-1}\ K^{-1}}$	
Avogadro constant	$N_A = 6.02 \times 10^{23}\ \mathrm{mol^{-1}}$	
acceleration due to gravity	$g = 9.81\ \mathrm{m\ s^{-2}}$ (close to the Earth)	
universal gravitational constant	$G = 6.67 \times 10^{-11}\ \mathrm{N\ m^2\ kg^{-2}}$	
speed of light in vacuum	$c = 3.00 \times 10^8\ \mathrm{m\ s^{-1}}$	
charge of electron	$e = 1.60 \times 10^{-19}\ \mathrm{C}$	
electron rest mass	$m_e = 9.11 \times 10^{-31}\ \mathrm{kg}$	
permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ \mathrm{C^2\ N^{-1}\ m^{-2}}$	
permeability of free space	$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H\ m^{-1}}$	
atomic mass unit	$\mathrm{u} = 1.661 \times 10^{-27}\ \mathrm{kg}$	(1 u is equivalent to 931 MeV)
astronomical unit	$\mathrm{AU} = 1.50 \times 10^{11}\ \mathrm{m}$	
light year	$\mathrm{ly} = 9.46 \times 10^{15}\ \mathrm{m}$	
parsec	$\mathrm{pc} = 3.09 \times 10^{16}\ \mathrm{m} = 3.26\ \mathrm{ly} = 206265\ \mathrm{AU}$	
Stefan constant	$\sigma = 5.67 \times 10^{-8}\ \mathrm{W\ m^{-2}\ K^{-4}}$	
Planck constant	$h = 6.63 \times 10^{-34}\ \mathrm{J\ s}$	

Rectilinear motion

For uniformly accelerated motion :

$$v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

Mathematics

Equation of a straight line	$y = mx + c$
Arc length	$= r\theta$
Surface area of cylinder	$= 2\pi rh + 2\pi r^2$
Volume of cylinder	$= \pi r^2 h$
Surface area of sphere	$= 4\pi r^2$
Volume of sphere	$= \frac{4}{3}\pi r^3$

For small angles, $\sin\theta \approx \tan\theta \approx \theta$ (in radians)

Astronomy and Space Science		Energy and Use of Energy	
$U = -\frac{GMm}{r}$	gravitational potential energy	$E = \frac{\Phi}{A}$	illuminance
$P = \sigma AT^4$	Stefan's law	$\frac{Q}{t} = \kappa\frac{A(T_H - T_C)}{d}$	rate of energy transfer by conduction
$\left\lvert\frac{\Delta f}{f_0}\right\rvert \approx \frac{v}{c} \approx \left\lvert\frac{\Delta\lambda}{\lambda_0}\right\rvert$	Doppler effect	$U = \frac{\kappa}{d}$	thermal transmittance U-value
		$P = \frac{1}{2}\rho Av^3$	maximum power by wind turbine

Atomic World		Medical Physics	
$\frac{1}{2}m_e v_{max}{}^2 = hf - \phi$	Einstein's photoelectric equation	$\theta \approx \frac{1.22\lambda}{d}$	Rayleigh criterion (resolving power)
$E_n = -\frac{1}{n^2}\left\{\frac{m_e e^4}{8h^2\varepsilon_0^2}\right\} = -\frac{13.6}{n^2}\ \mathrm{eV}$	energy level equation for hydrogen atom	$\text{power} = \frac{1}{f}$	power of a lens
		$L = 10\log\frac{I}{I_0}$	intensity level (dB)
$\lambda = \frac{h}{p} = \frac{h}{mv}$	de Broglie formula	$Z = \rho c$	acoustic impedance
		$\alpha = \frac{I_r}{I_0} = \frac{(Z_2 - Z_1)^2}{(Z_2 + Z_1)^2}$	intensity reflection coefficient
$\theta \approx \frac{1.22\lambda}{d}$	Rayleigh criterion (resolving power)	$I = I_0 e^{-\mu x}$	transmitted intensity through a medium

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(b) Now the coil is rotated uniformly about an axis through 180° as shown in Figures 9.2(a) and 9.2(b) within 0.5 s.

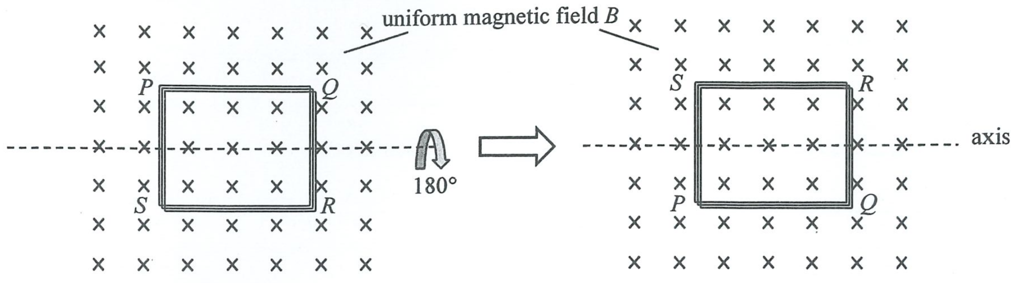


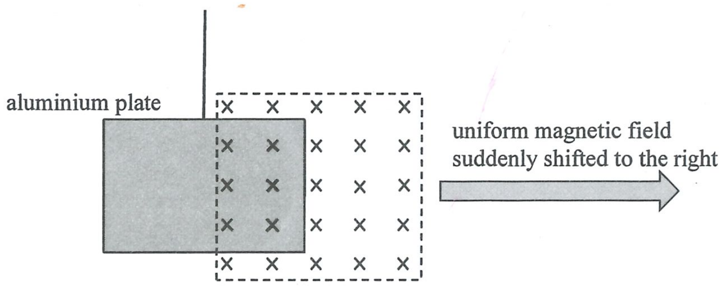
Figure 9.2(a) Figure 9.2(b)

(i) State the value of the change in total magnetic flux linkage through the coil in this case (1 mark)

(ii) At the moment when the coil rotated through 90°, would the induced current flow in the direction *PQRS*, *PSRQ* or is there no induced current in the coil ? (1 mark)

(c) Figure 9.3 shows a thin rectangular aluminium plate suspended by a long string. The plate is partly inside a uniform magnetic field provided by a strong magnet.

Figure 9.3



The magnet, which is not in contact with the plate, is suddenly shifted to the right.

(i) On Figure 9.3, draw a small circle at the location where eddy currents are induced on the aluminium plate. Use an arrow to indicate the direction of current. (2 marks)

(ii) Describe the subsequent motion of the aluminium plate, if any. (1 mark)

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10. (a) The equation below represents nuclear fission of uranium-235 (U-235).

$$^{235}_{92}\text{U} + {}^{1}_{0}\text{n} \rightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + x\,{}^{1}_{0}\text{n} + 200\text{ MeV}$$

(i) What is the value of x ? (1 mark)

(ii) State a necessary condition for chain reaction of fission to occur. (1 mark)

Scientists found evidence in Oklo, Africa that natural nuclear fission occurred two billion (2×10^9) years ago. The uranium mineral ore mined from Oklo **at present** is found to have 0.6% concentration by mass of U-235 (see the table below), which is much lower than usual.

(b) The table gives the information of U-235 and U-238 in a sample of uranium mineral ore found in Oklo. Given: half-life of U-235 = 7.04×10^8 years

	2×10^9 years ago	**at present**
U-235	m_0 kg	0.060 kg (i.e. 0.6% concentration by mass)
U-238	13.556 kg	9.940 kg (i.e. 99.4% concentration by mass)

*(i) Estimate the amount m_0 (in kg) of U-235 in the sample 2×10^9 years ago. (2 marks)

(ii) Hence determine whether natural nuclear fission of U-235 was possible 2×10^9 years ago. For fission of U-235 to happen, its concentration by mass in the uranium mineral ore has to be at least 3%. (1 mark)

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There must be underground water in the vicinity of this uranium-rich mineral deposit for natural nuclear fission to be possible. Since water can slow down the fast neutrons from fission, these neutrons can easily be captured by U-235.

(c) In fact the chain reaction stopped even before the concentration by mass of U-235 dropped to 3%. Explain why this occurred. (2 marks)

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(b) The potential difference across A and B is 5.0 kV and the separation between the metal plates is 10 cm. The force due to the electric field acting on each sphere is 2.0×10^{-5} N, find

(i) the moment acting on the rod as shown in Figure 9.2 due to the electric forces on the charged spheres. (2 marks)

*(ii) the strength of the electric field E due to the potential difference across the metal plates. (2 marks)

(iii) the magnitude of the charge Q on the spheres. (2 marks)

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10. Given: mass of proton = 1.0073 u
mass of α particle = 4.0015 u
mass of $^{14}_{7}N$ nucleus = 13.9993 u
mass of $^{17}_{8}O$ nucleus = 16.9947 u

When a stationary $^{14}_{7}N$ nucleus is bombarded by an α particle, the following nuclear reaction can be triggered with products $^{17}_{8}O$ and X fly off:

$$\alpha + {}^{14}_{7}N \longrightarrow {}^{17}_{8}O + X$$

(a) What is X? (1 mark)

*(b) Based on energy consideration, estimate the minimum kinetic energy, in MeV, of the α particle required for such a nuclear reaction to occur. (2 marks)

(c) However, when conservation of momentum is also taken into account, the α particle must possess a kinetic energy greater than that found in (b) to bring about such a reaction. Explain. (2 marks)

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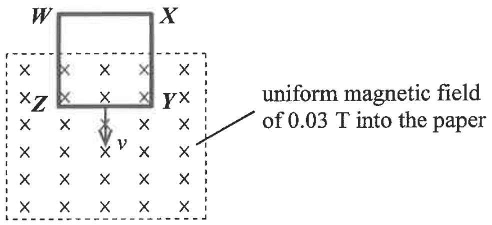
9. A square metal loop *WXYZ* of side length 0.10 m is inserted with a constant velocity v into a uniform magnetic field of flux density 0.03 T. The magnetic field is perpendicular to the plane of the loop as shown in Figure 9.1. The resistance of **each side** of the metal loop is 0.15 Ω.

Figure 9.1



When the metal loop is entering the magnetic field, a current of 0.01 A flows in the loop.

(a) On Figure 9.1, indicate the direction of this current. (1 mark)

*(b) Find v. (2 marks)

(c) (i) Find the **potential difference** V_{YZ} between Y and Z. (2 marks)

(ii) Explain whether V_{YZ} is equal to the **induced e.m.f.** across YZ. (1 mark)

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10. Deuterium (^2_1H) and tritium (^3_1H) are isotopes of hydrogen. The fusion of two deuterium nuclides can be represented by the following equation:

$$^2_1H + ^2_1H \longrightarrow ^3_2He + ^1_0n$$

Given: mass of ^2_1H = 2.014102 u
mass of ^3_2He = 3.016029 u
mass of ^1_0n = 1.008665 u

*(a) 1 in 6420 atoms of hydrogen in nature is a deuterium. Estimate the maximum energy, in MeV, that could be produced from this fusion reaction with 1 mole of hydrogen nuclides. (3 marks)

(b) If conditions are altered, the fusion of two deuterium nuclides may **not** produce a helium (He) nucleus. Complete the equation below for such a possible fusion reaction. (1 mark)

$$^2_1H + ^2_1H \longrightarrow \dots\dots + \dots\dots$$

(c) Both fission and fusion produce energy. State **TWO** advantages of using fusion as an energy source over fission. (2 marks)

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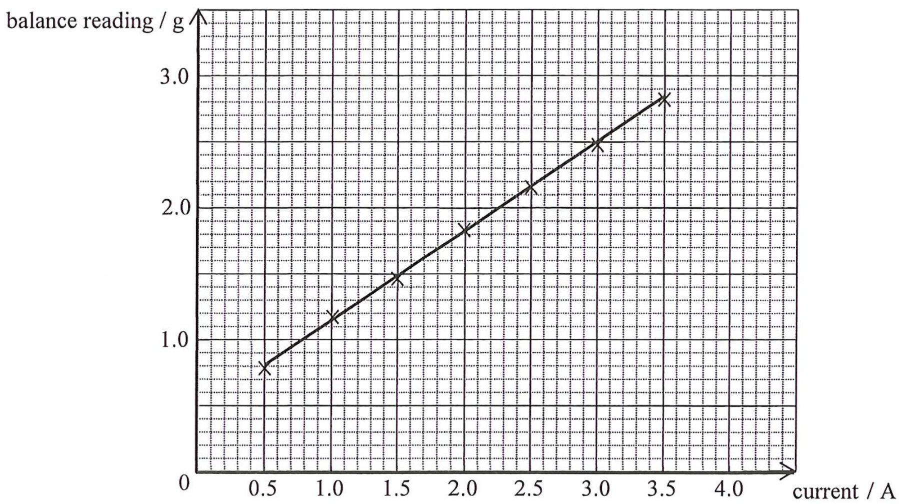
(d) The graph below is plotted by using the experimental data obtained.



(i) Describe the corresponding experimental procedures to be carried out. (2 marks)

(ii) Describe how the balance reading varies with the current as revealed by the graph. (1 mark)

(iii) Explain why the graph does not pass through the origin when the current is zero. (1 mark)

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9. Part of the decay series of thorium-232 (Th-232) is shown below. Th-232 decays to radium-228 (Ra-228) with a half-life of 1.41×10^{10} years. The end product lead-208 (Pb-208) is stable.

$$^{232}_{90}\text{Th} \longrightarrow {}^{228}_{88}\text{Ra} \xrightarrow{\beta} \text{Ac} \rightarrow \ldots \rightarrow {}^{208}_{82}\text{Pb}$$

(a) (i) State the kind of radiation emitted by Th-232. (1 mark)

(ii) How many α- and β- decay(s) has occurred before a Th-232 nucleus becomes a Pb-208 nucleus ? (2 marks)

*(b) Estimate the proportion of Th-232 decaying in 10 years. Give your answer correct to 2 significant figures. (2 marks)

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(c) By adding thorium oxide (a compound containing Th-232) to glass, thoriated glass having a higher refractive index is produced. Camera lenses manufactured before 1970 are often made from thoriated glass.

Figure 9.1

(i) Although thoriated glass is radioactive, practically it is safe to use a camera with such lenses like the one shown in Figure 9.1. Why ? (1 mark)

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(ii) According to the decay series of thorium-232, thorium oxide would end up as lead oxide which is black in colour. Why is there no need to worry that a thoriated glass lens would turn black when used for several years ? (1 mark)

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(i) Draw an arrow in the figure below to represent the direction of the acceleration of the block at that moment. (1 mark)

(ii) Find the horizontal displacement of the block from that moment until just before it reaches the ground. (3 marks)

(iii) Describe the energy conversion of the block from that moment until just before it reaches the ground. (2 marks)

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(i) Write the nuclear equations corresponding to each of these two decay processes. (2 marks)

*(ii) The half-life of Bi-213 is 46 minutes. At time $t = 0$ s, its number of undecayed nuclei is 8.0×10^{11}. Find its activity (in Bq) at $t = 0$ s. (2 marks)

*(iii) Find the percentage decrease in the activity after 200 minutes. (2 marks)

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Figure 12.2(a) Figure 12.2(b)

(i) Compared to placing it in a badge worn on the chest, what is the advantage of placing the chip in the ring worn on the hand ? (1 mark)

(ii) The ring should be worn under the gloves. Explain briefly. (1 mark)

(iii) Which type(s) of radiation (α, β or γ) can reach the chip of the ring ? (1 mark)

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